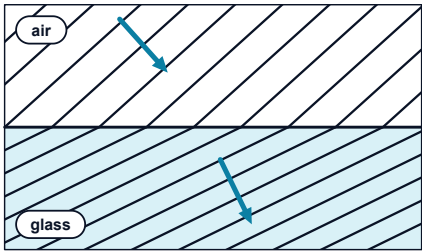
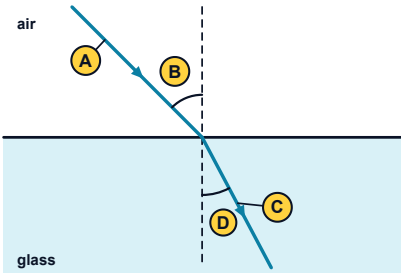


Electromagnetic waves

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 14 starred (★) questions first.

	QUESTION	ANSWER	✓
1	Light passes from air into glass. Compare the angle of refraction with the angle of incidence.	It is smaller (the ray bends towards the normal).	☐
2	Light passes from glass into air. Compare the angle of refraction with the angle of incidence.	It is larger (the ray bends away from the normal).	☐
3 ★	What type of wave are electromagnetic waves?	Transverse	☐
4	State what oscillates in electromagnetic waves.	Electric and magnetic fields	☐
5	Explain why electromagnetic waves can travel in a vacuum.	They are oscillations of fields, not particles, so they need no medium.	☐
6 ★	Name the seven groups of the electromagnetic spectrum, from longest to shortest wavelength.	Radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays	☐
7	Name the electromagnetic waves with the longest wavelength.	Radio waves	☐
8	Name the electromagnetic waves with the shortest wavelength.	Gamma rays	☐
9	Name the electromagnetic waves that human eyes can detect.	Visible light	☐
10 ★	Define refraction.	The change in direction of a wave at a boundary between two media.	☐
11 ★	Give two situations in which a wave crossing a boundary is not refracted.	When it travels along the normal. When its speed is the same in both media.	☐
12 ★	Explain why refraction happens.	The wave changes speed in the new medium.	☐

QUESTION	ANSWER	✓
13 Name the process shown in this wavefront diagram. 	Refraction	☐
14 State what labels A to D show in this diagram. 	A: incident ray B: angle of incidence C: refracted ray D: angle of refraction	☐
15 ★ Describe how radio waves can be produced.	By oscillations in an electrical circuit.	☐
16 ★ Describe how radio waves can be detected.	They are absorbed by an aerial, creating an alternating current of the same frequency.	☐
17 ★ State two hazards of ultraviolet radiation.	Premature skin ageing and a higher risk of skin cancer	☐
18 ★ State the hazards of X-rays and gamma rays.	They are ionising: they can cause gene mutations and cancer.	☐
19 State what radio waves are used for.	Television and radio signals	☐
20 State what microwaves are used for.	Satellite communications and cooking food	☐
21 ★ State what infrared radiation is used for.	Electrical heaters, cooking food and infrared cameras	☐
22 State what visible light is used for.	Fibre optic communications	☐
23 ★ State what ultraviolet radiation is used for.	Energy-efficient lamps and sun tanning	☐
24 State what X-rays and gamma rays are used for.	Medical imaging and treatments	☐

QUESTION		ANSWER	✓
25 ★	State what all electromagnetic waves have in common in a vacuum.	They travel at the same speed.	☐
26	Name the electromagnetic waves with the highest frequency.	Gamma rays	☐
27	State what electromagnetic waves transfer.	Energy, from a source to an absorber.	☐
28	State the kind of spectrum that electromagnetic waves form.	A continuous spectrum	☐
29	State where gamma rays originate.	From changes in the nucleus of an atom.	☐
30 ★	Explain why microwaves are suitable for satellite communications.	They pass through the atmosphere without being absorbed or reflected.	☐
31 ★	Explain why X-rays are suitable for producing images of bones.	They pass through soft tissue but are absorbed by bone.	☐
32	State two things that the hazard from UV, X-rays and gamma rays depends on.	The type of radiation and the size of the dose	☐